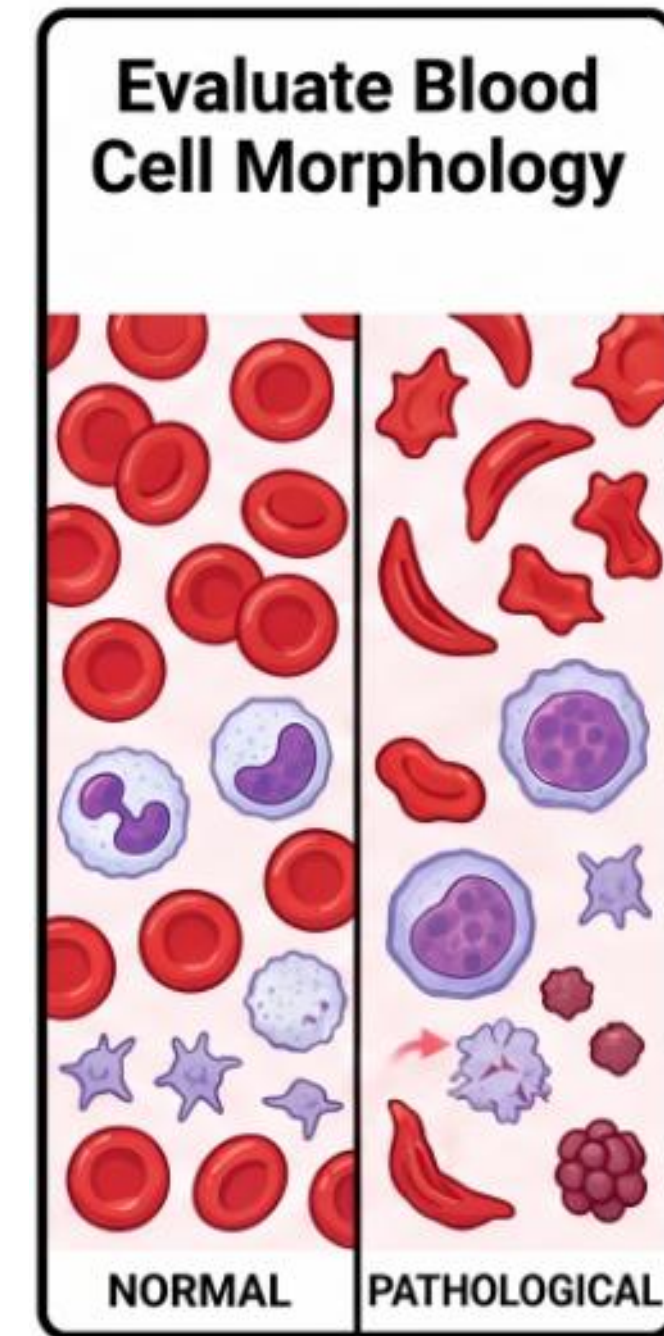
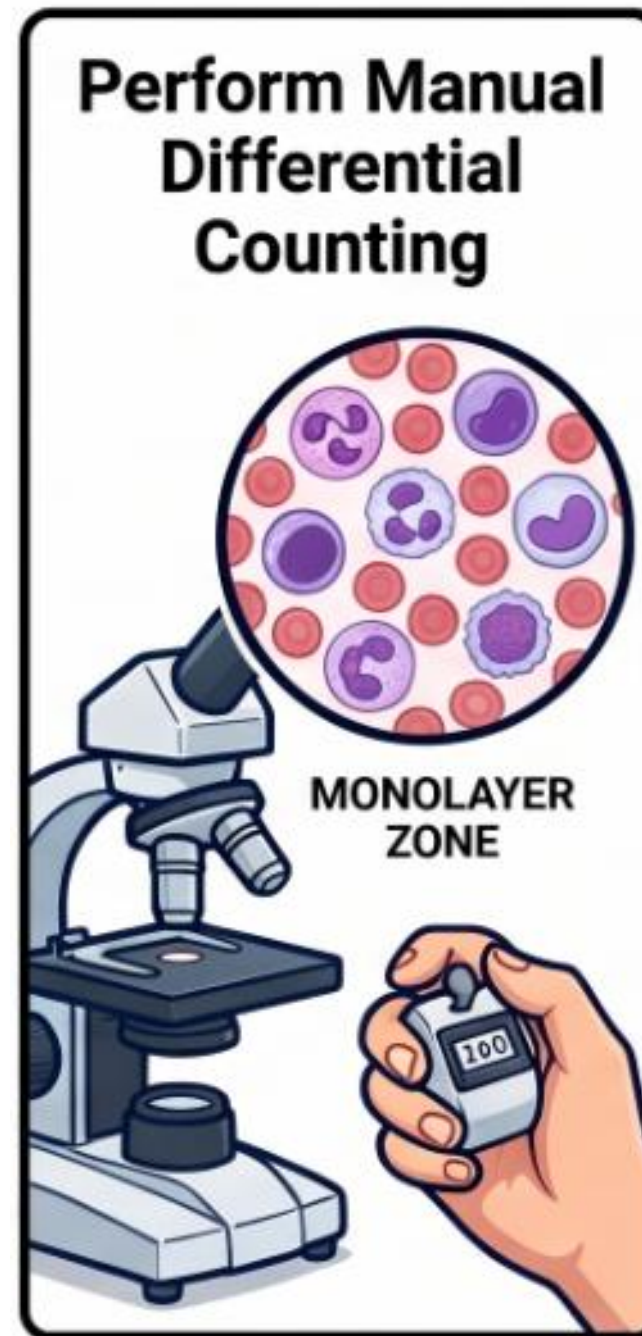
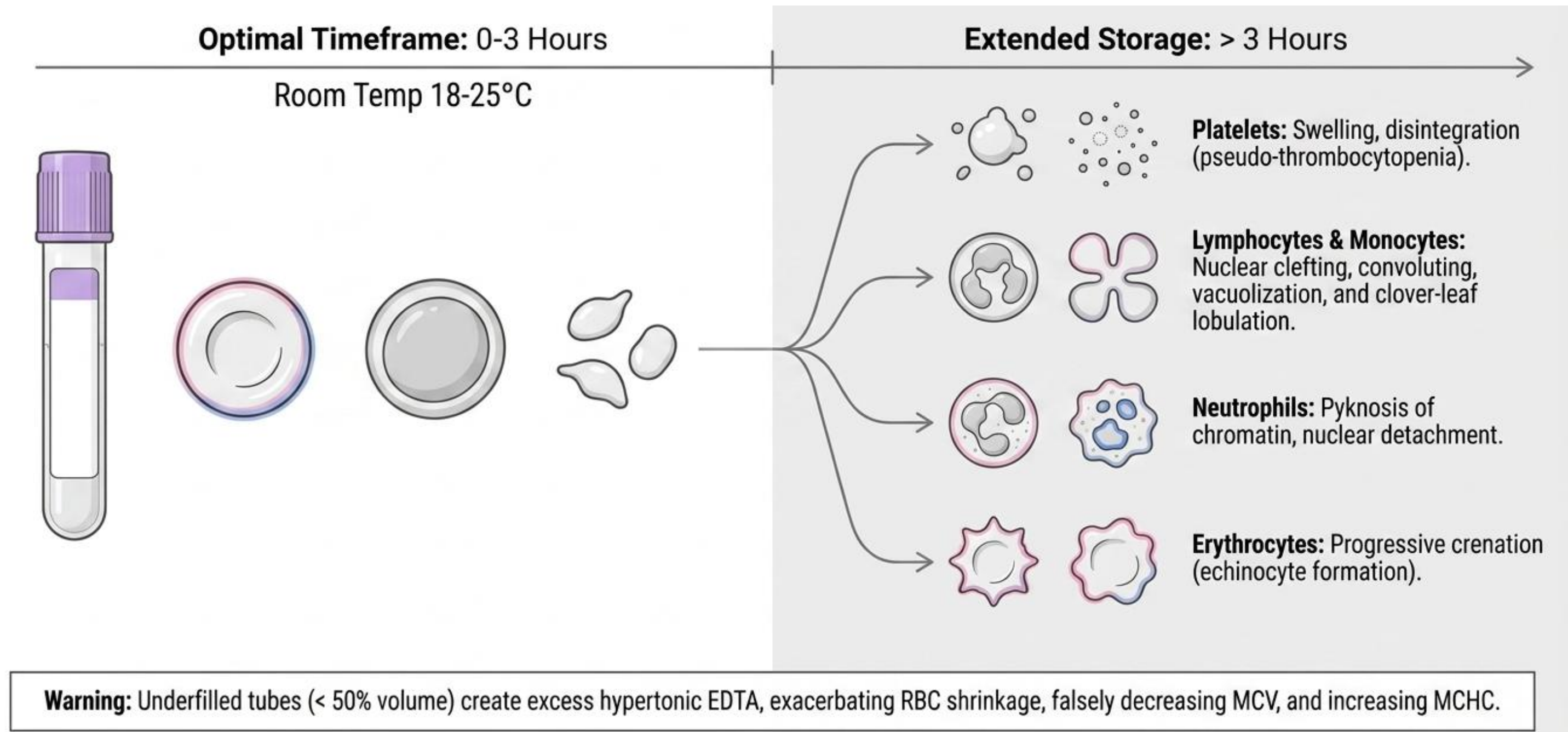


PBS Preparation

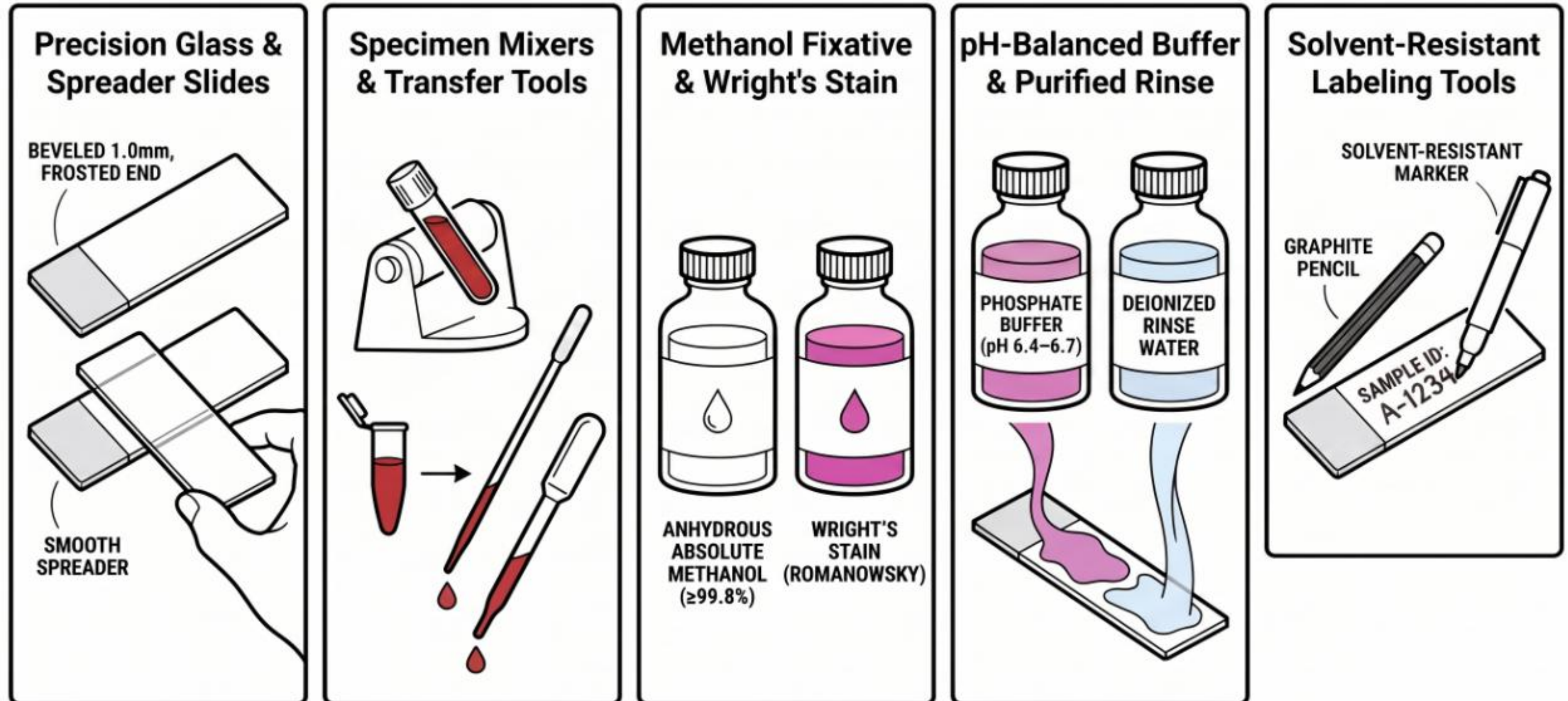
Clinical Significance



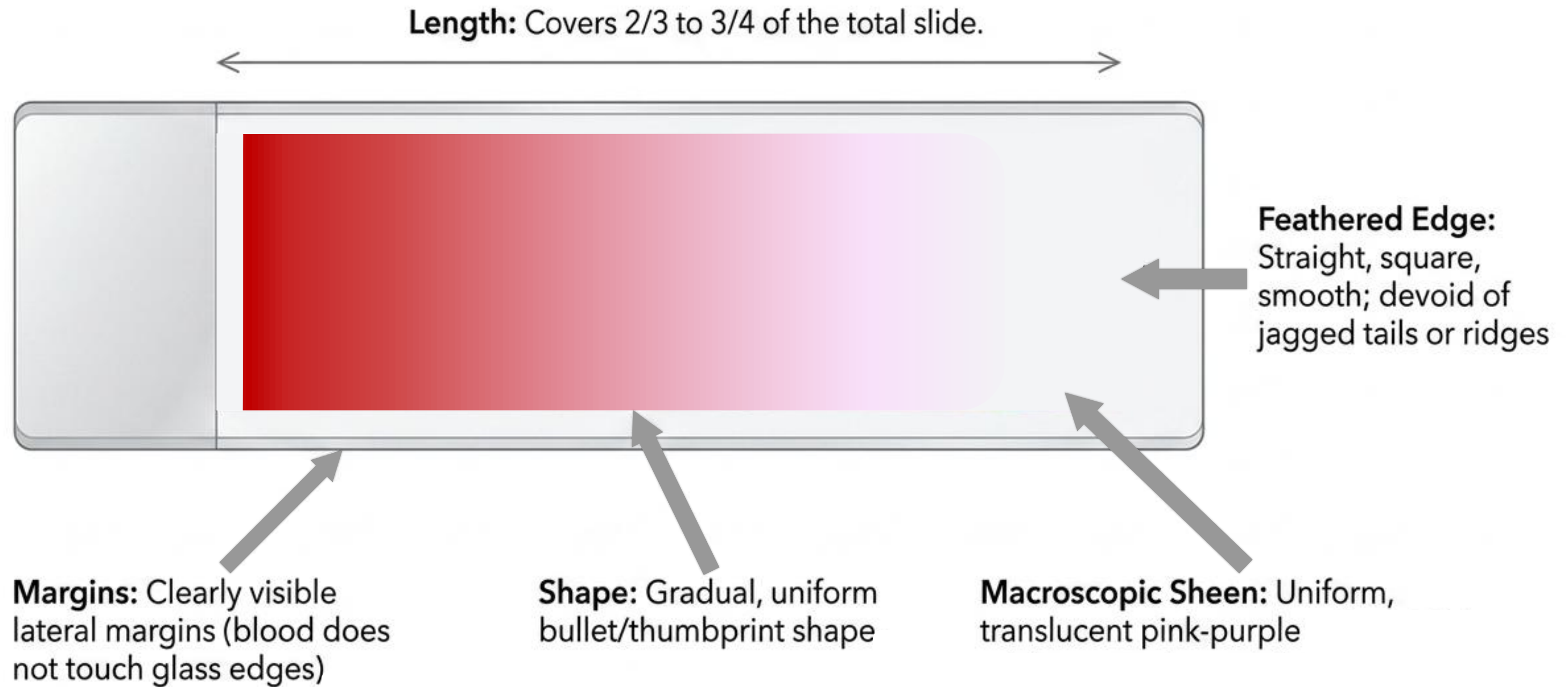
Pre-Analytical & Specimen Integrity



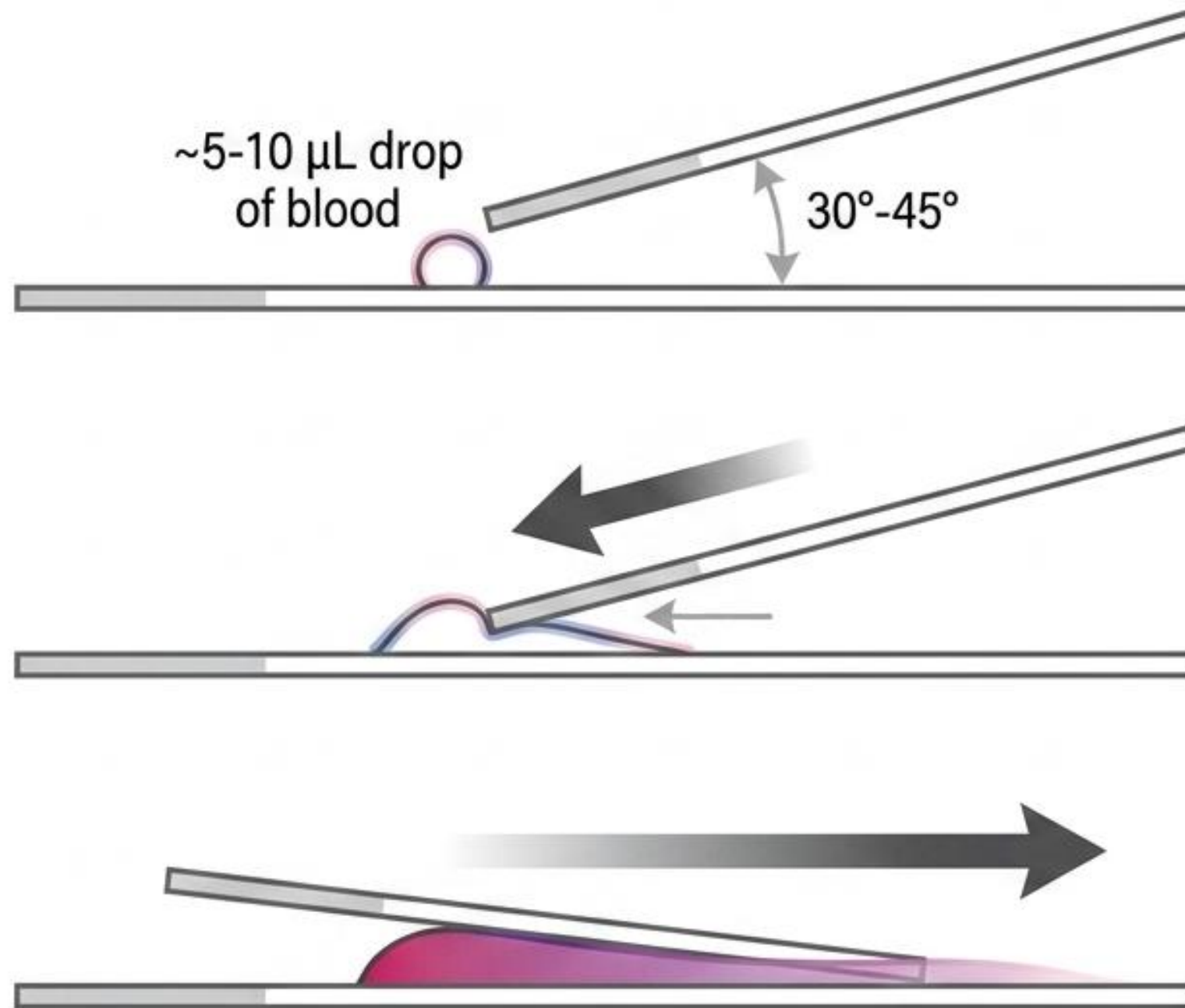
Equipment, Supplies & Reagents



Ideal Smear



Push-Wedge Mechanics



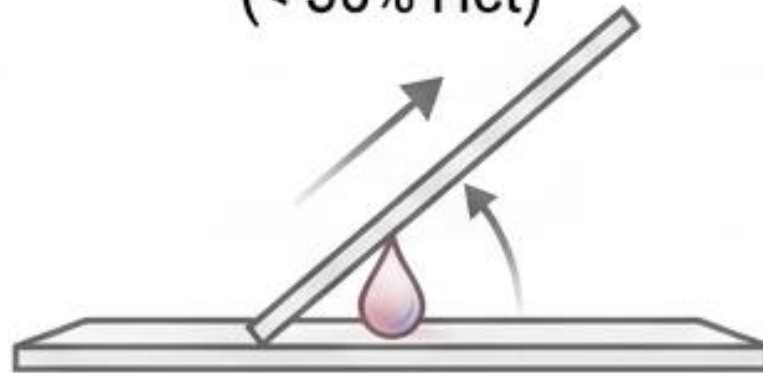
1. Dispense & Position: Place a ~5-10 μL drop of blood 1/2 inch from the frosted end. Hold spreader slide ahead of the drop at exactly a 30°-45° angle.

2. Back-Up & Spread: Draw the spreader backward into the drop. Allow blood to spread along the capillary contact line to within 1-2 mm of the edges.

3. The Forward Stroke: Rapid, smooth, continuous motion. Do not lift or alter downward pressure.

Compensating for Hematocrit

Severe Anemia ($< 30\%$ Hct)



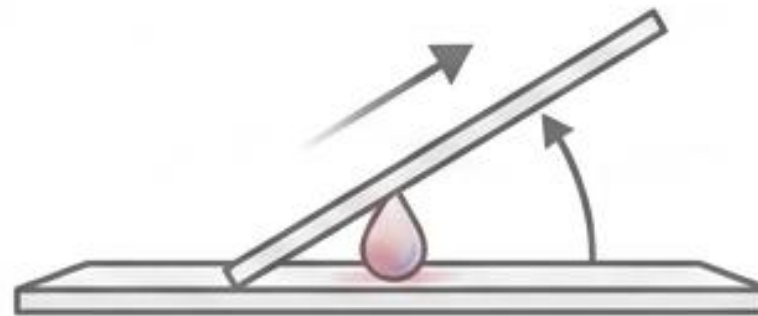
Challenge: Smear overly long/thin; cells run off edge.

Physics Adjust: Steep Angle ($> 45^\circ$) + Fast Speed.

Result: Forces a thicker, shorter film to maintain cell density.



Normal Blood ($36\text{-}50\%$ Hct)

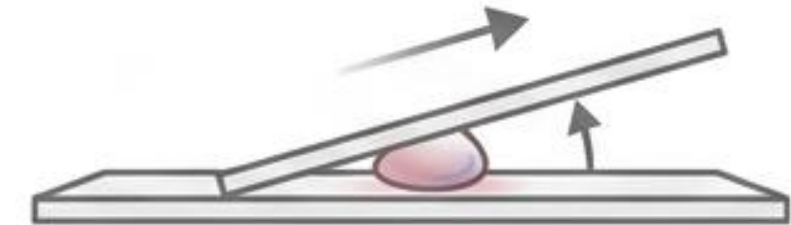


Physics Adjust: Standard Angle ($30^\circ\text{-}45^\circ$) + Moderate Speed.

Result: Optimal $2/3$ length monolayer.



Polycythemia / Neonates ($> 55\%$ Hct)



Challenge: Smear thick, crowded; lacks monolayer.

Physics Adjust: Flat Angle ($< 30^\circ$) + Slow Speed.

Result: Thinner, extended film allowing cell separation.



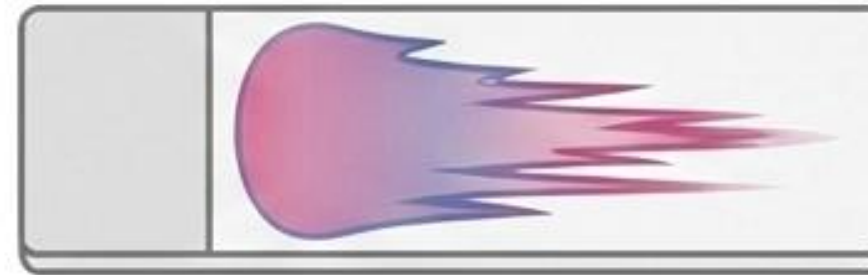
Common Technical Smear Defects



Washboard Ridges

Cause: Hesitation / jerky forward motion.

Microscopic: Cells clumped in ridges.



Longitudinal Streaks / Tails

Cause: Chipped spreader slide edge.

Microscopic: Falsely high WBCs dragged into tails.



Drop Too Large

Microscopic: No readable monolayer; cell overlap.

Drop Too Small



Microscopic: Insufficient monolayer area.

Macroscopic Clumping at Tail



Cause: Delayed spread after drop placement.

Microscopic: Rouleaux, platelet clumps.

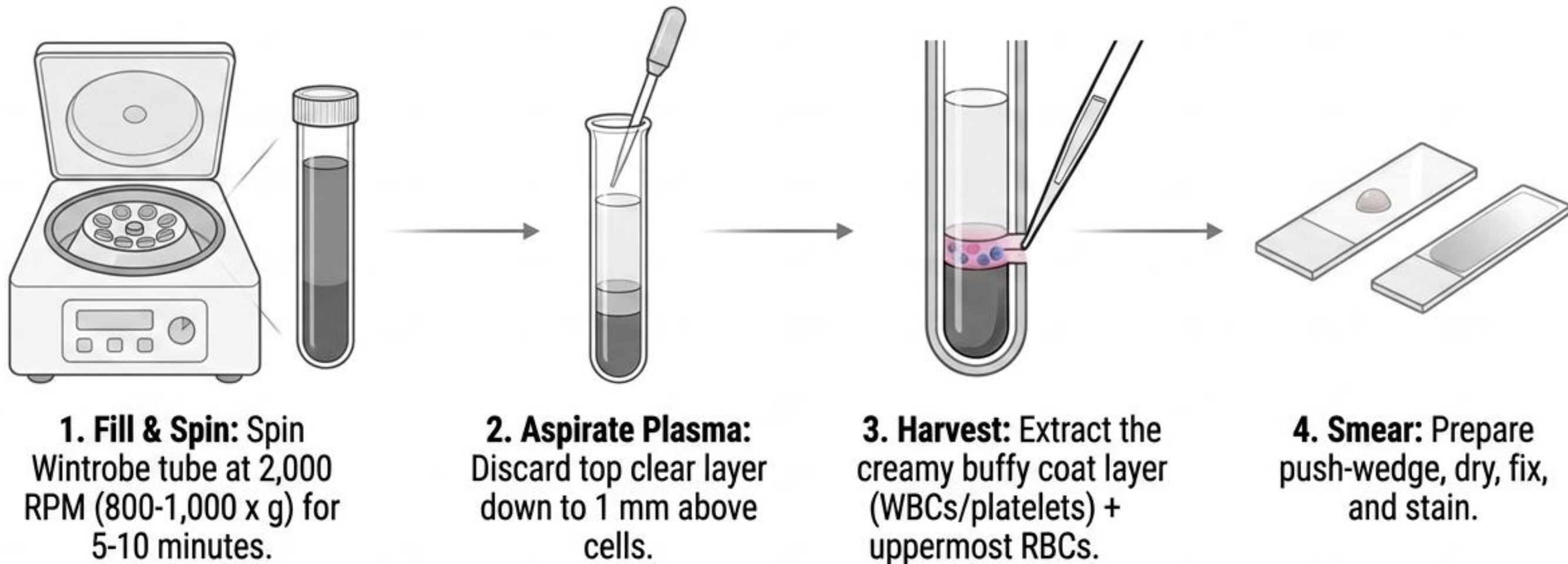
Scratched Glass / Pale Streaks



Cause: Excess downward pressure.
Microscopic: Crushed leukocytes (smudge cells).

Special Technique: Buffy Coat Blood Smears

Special Technique: Extreme Leukopenia ($\text{WBC} < 2.0 \times 10^3/\mu\text{L}$)



Clinical Warning: Differential is qualitative. Evaluate WBC morphology and blasts. RBC morphology must NEVER be reported from buffy coat smears.

Romanowsky Staining Chemistry

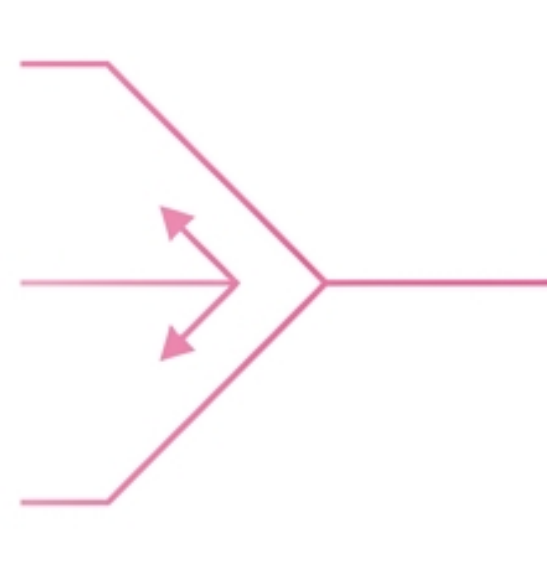
Base Solvent: Pure Methanol (dehydrating chemical fixative)

Methylene Blue⁺ + Eosin Y⁻

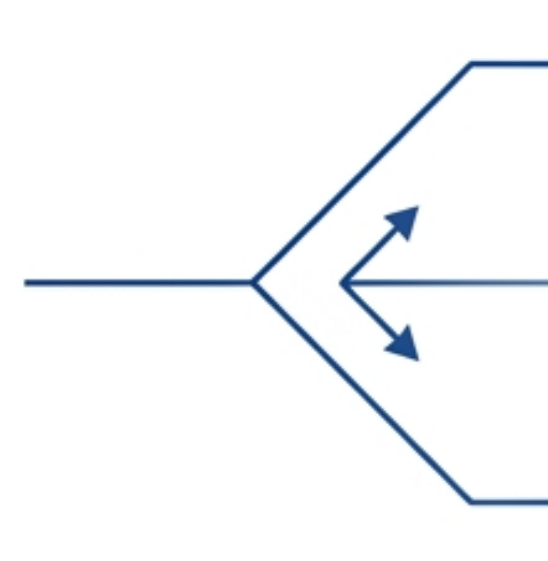
Eosin Y/B

Charge: Anionic / Acidic
Dye (Dye⁻)

Affinity: Binds to basic
(acidophilic/eosinophilic)
cellular polycations.



+



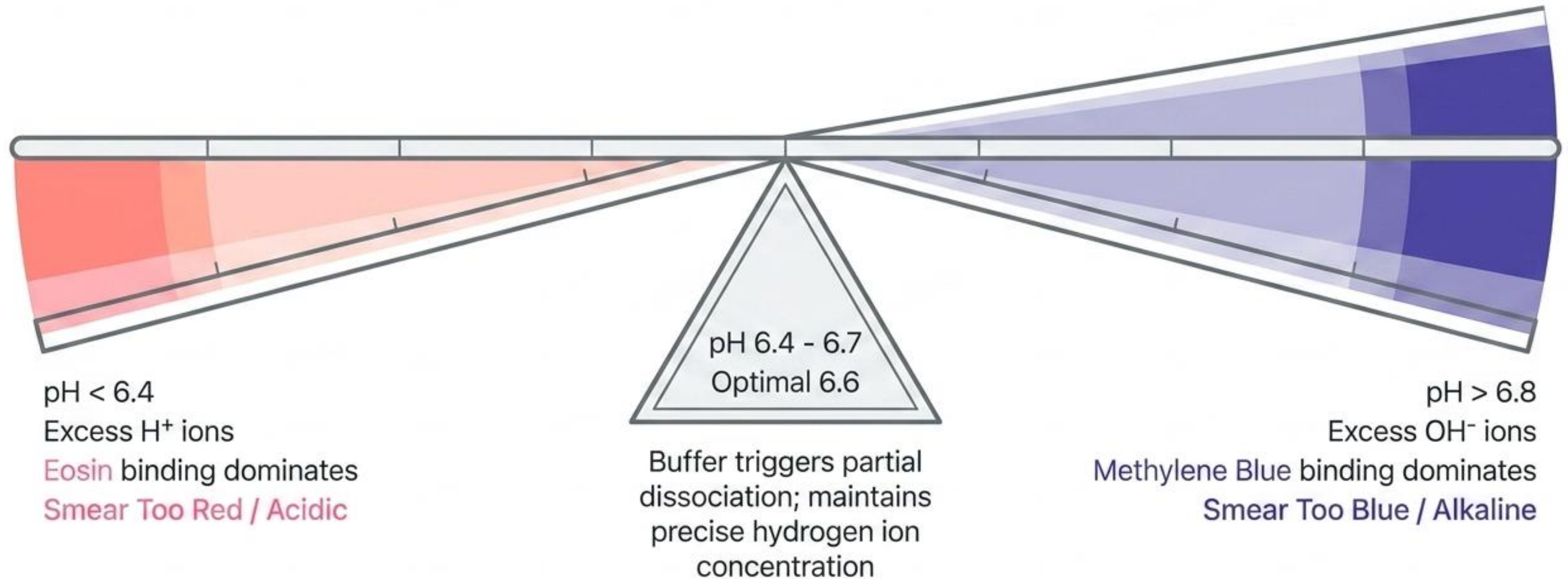
Methylene Blue/Azure

Charge: Cationic / Basic
Dye (Dye⁺)

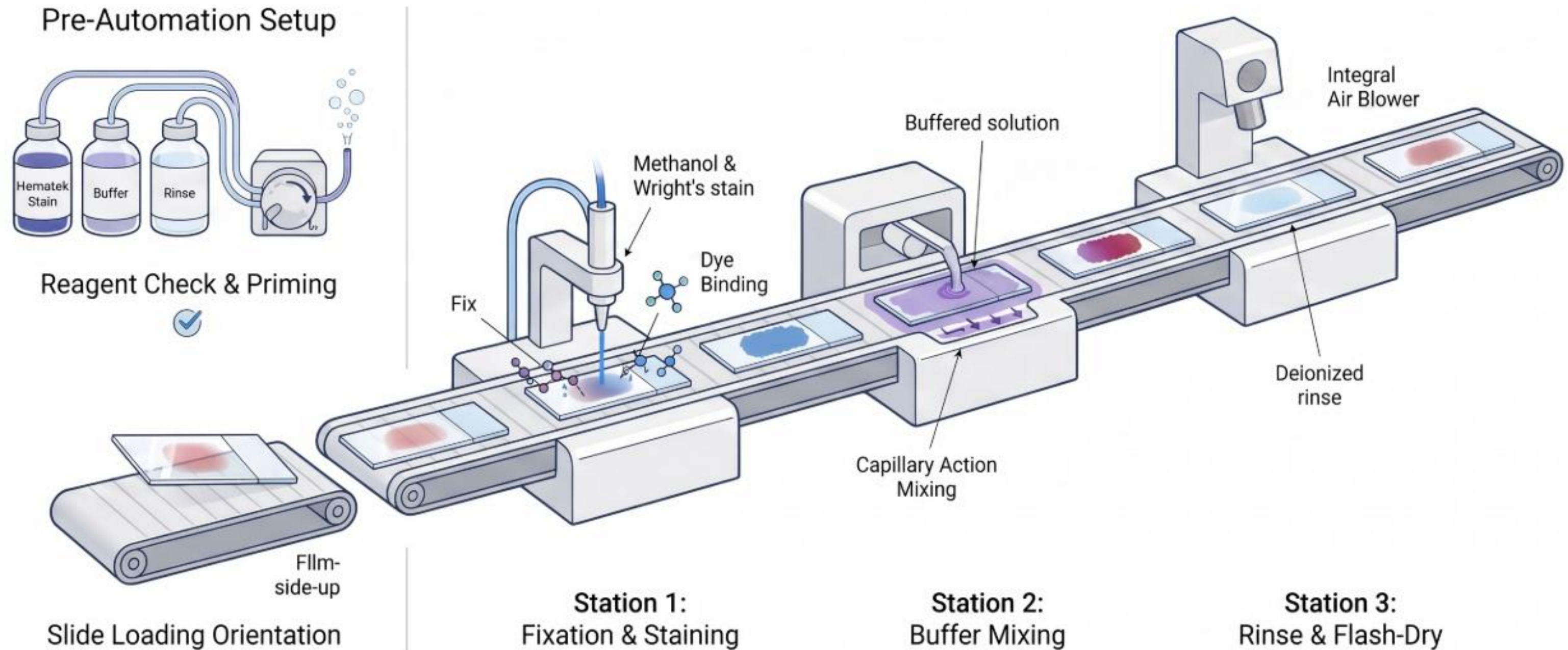
Affinity: Binds to acidic
(basophilic) cellular
polyanions.

Undissociated complex stains
neutral structures lilac-pink.

Role of Buffer & pH Control



Automated Staining Protocol

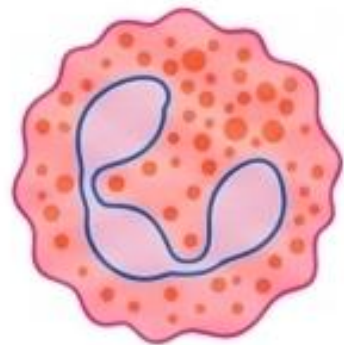


Expected Microscopic Staining Characteristics

Pink Targets (Basic Proteins binding Eosin Y)



Erythrocytes (RBCs):
Pink to salmon-buff
(Target: Hemoglobin)



Eosinophil Granules:
Bright red-orange, refractive
(Target: Basic cationic proteins)

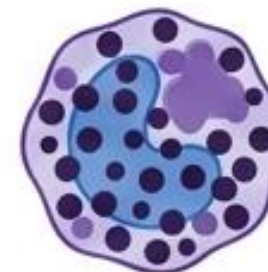
Blue/Purple Targets (Acidic Polyanions binding Methylene Blue)



Neutrophil Nuclei: Deep blue-purple
(Target: Nucleic acid phosphates/DNA)



Lymphocyte Cytoplasm: Robin's egg blue
(Target: Ribosomal RNA)



Basophil Granules:
Dark purple-black, water-soluble
(Target: Heparin/Histamine)

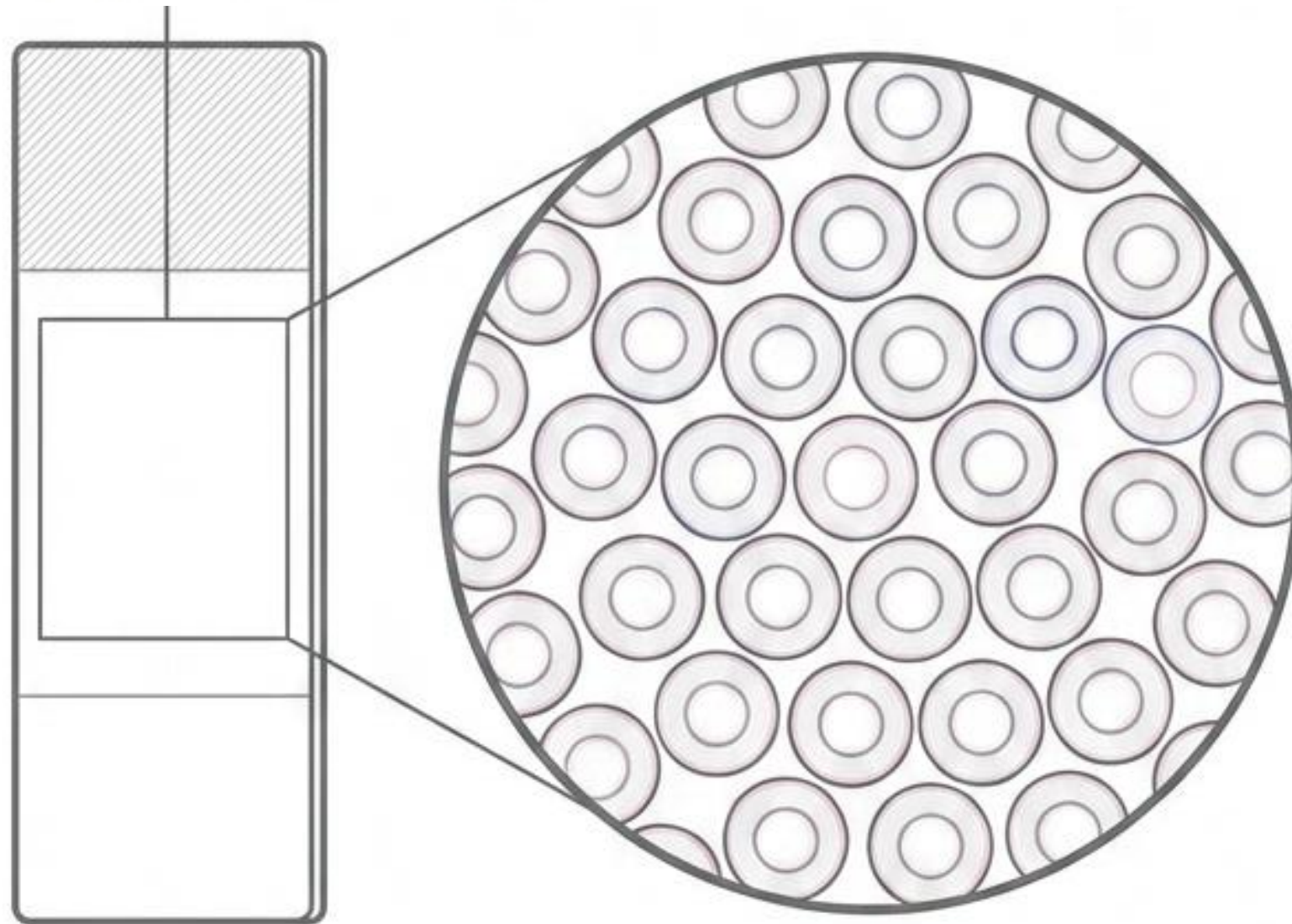
Neutral Zone: Neutrophil cytoplasm appears pale pink/lilac with fine pink granules
(Target: Neutral protein complexes)



Platelets: Violet-purple
(Target: Glycoprotein matrix)

Smear Evaluation: Monolayer Zone Selection

Zone of Optimum Morphology



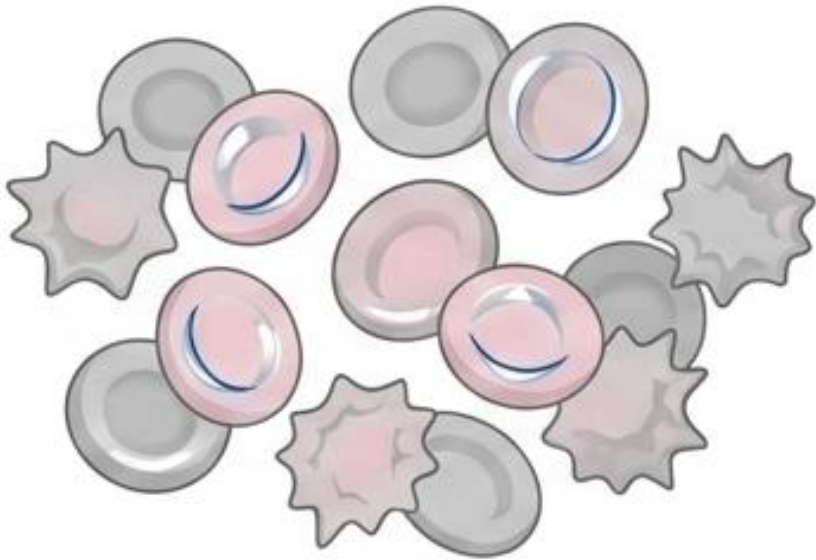
Microscopic Monolayer Criteria
(100x Oil Immersion)

1. RBC Distribution: Evenly spaced, single layer, barely touching.
2. Central Pallor: Clearly discernible, exactly $\frac{1}{3}$ the diameter of the cell.
3. Leukocyte Detail: Chromatin detail and cytoplasmic granules crisply resolved.
4. Platelets: Granule distribution distinctly visible.

Stain Troubleshooting

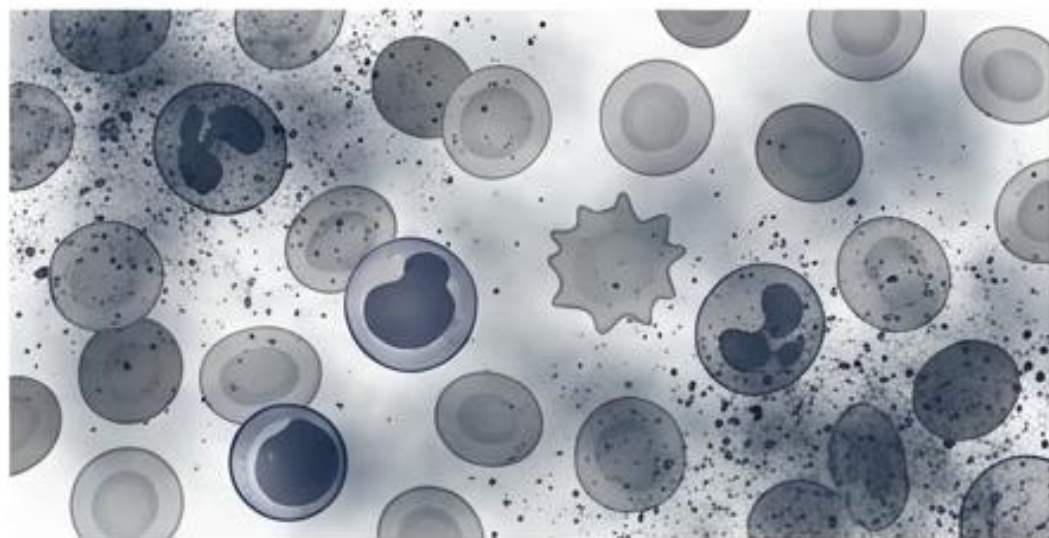
Too Acidic (Excessively Red)	Too Alkaline (Excessively Blue)
Macroscopic: Bright red/orange film.	Macroscopic: Dark blue/greenish-gray film.
Microscopic: Pale/washed-out nuclei; bright orange RBCs; intense bright yellow-red eos granules.	Microscopic: Dense purple-black nuclei; slate-green/gray RBCs; dark slate-blue neutrophil cytoplasm.
Root Causes: Low buffer pH, old/oxidized methanol stain (formic acid), over-washing, acid fumes.	Root Causes: Buffer pH > 6.8, prolonged staining, staining, inadequate rinse, excessively thick smear trapping dye, unaged stain (< 2-3 weeks).
Corrective Actions: Replace aged stain pack; recalibrate buffer to 6.60; increase stain time / decrease wash.	Corrective Actions: Adjust pH downward; shorten incubation; extend rinse; thin out smears.

Stain Artifacts



Water Artifact

- ☐ Visual Effect: Refractile, glassy rings inside RBCs and severe RBC crenation.
- ☐ Mechanism: Moisture contamination prior to complete methanol fixation.
- ☐ Root Causes: High ambient humidity; blowing warm breath on wet slide; water contamination in methanol.
- ☐ Fix: Absolute anhydrous methanol fixation before staining; mechanical fan drying only.



Stain Precipitate

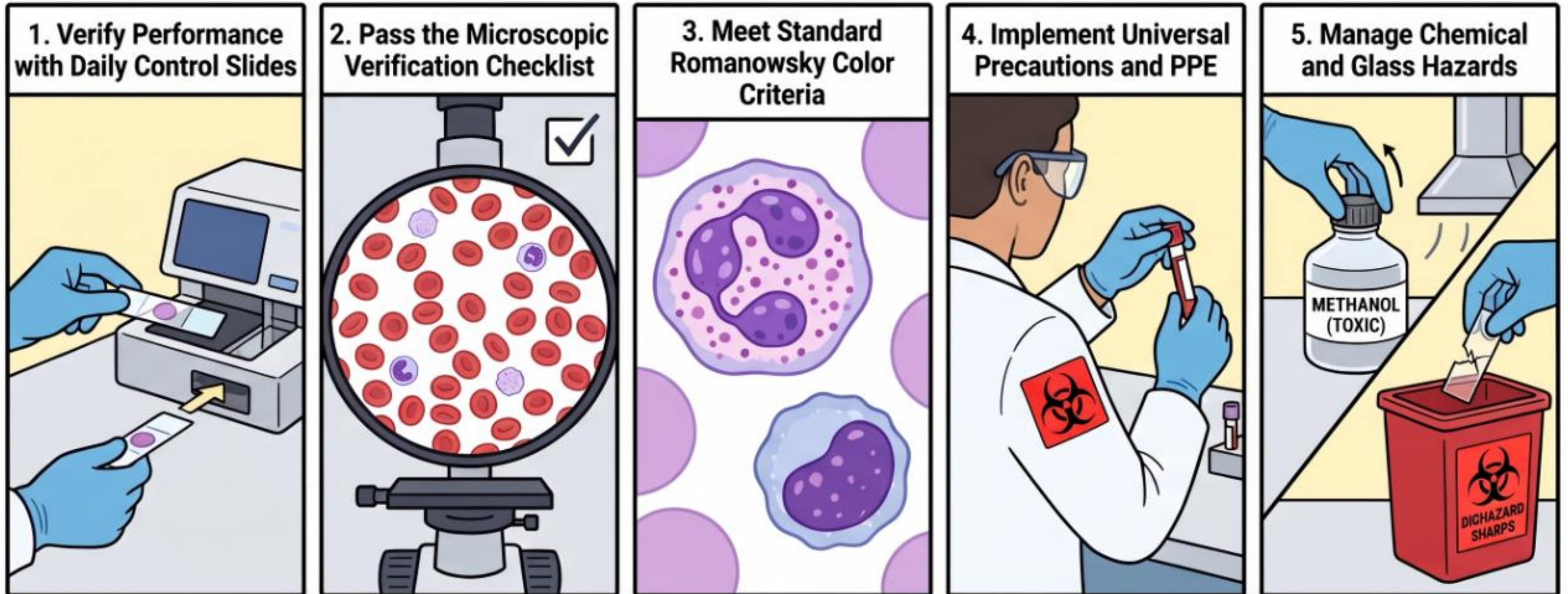
- ☐ Visual Effect: Amorphous dark blue/black grit or haze over the entire slide background and cells.
- ☐ Root Causes: Unfiltered/dirty stain solution; slides drying out mid-staining; inadequate post-stain rinse volume.
- ☐ Fix: Filter stain daily; ensure continuous reagent flooding; increase wash flow.

Special Stains

Supravital Stain (NMB / Brilliant Cresyl Blue)	Prussian Blue Reaction
Stains living, unfixed cells. Precipitates residual ribosomal RNA.	Acid ferrocyanide reaction. Detects stored Fe^{3+} (ferric iron).

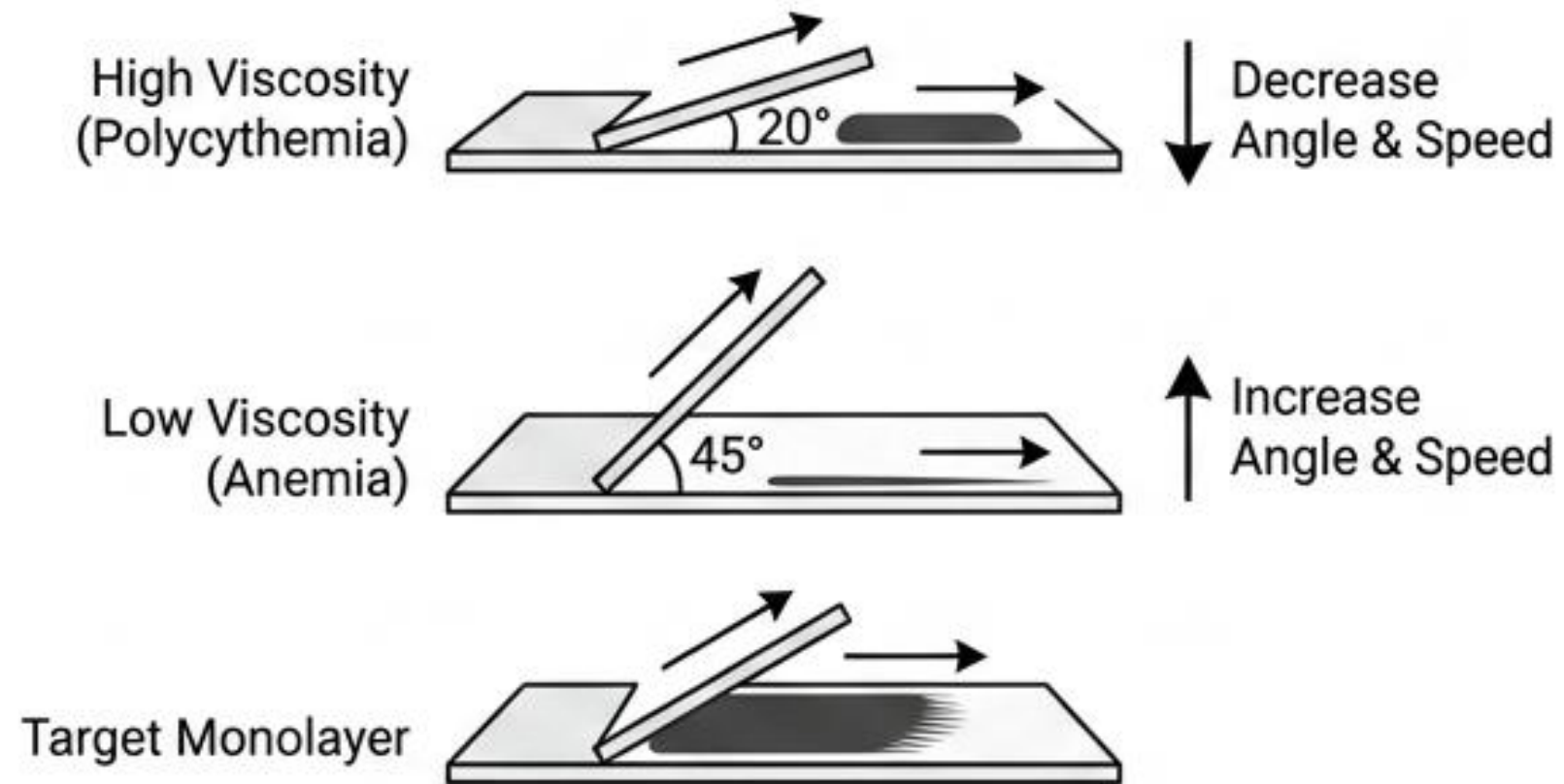
Inclusion	Wright's Stain	Prussian Blue	Composition
Pappenheimer Bodies	Pale blue dots	Positive (Bright blue)	Ferric iron
Howell-Jolly Bodies	Dark-purple dot	Negative	Nuclear DNA fragment
Basophilic Stippling	Diffuse blue dots	Negative	Ribosomal RNA
Heinz Bodies	Invisible	Requires NMB/Supravital	Denatured hemoglobin

Quality Control & Biosafety



Technical Summary & Key Takeaways

The Physics of the Push



The Chemistry of the Stain

